

Learned Latent Curves: A Closed-Loop Framework for Corpus Embedding and Recursive Skill Auditing

v2 — with the Hyperspherical-Harmonic-Curve Variant

Shant Tchatalbachian + Sauna (cycle 1–5 RSI)
foil-copy-overrate@duck.com | yubiOS

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Abstract. We describe a family of composed techniques used to audit and prioritize improvement work across a corpus of engineering artifacts (software skills, self-documentation files): (1) a learned latent curve, which re-expands a single 1-D ordering coordinate t into a fixed-width D -dimensional embedding via a Fourier basis with learned frequencies; (2) curve-guided recursive self-improvement (curve-guided RSI), which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; (3) a self-documentation variant of the same pipeline, retargeted at heterogeneous memory-file corpora via a per-corpus primitive basis and an explicit item-granularity rule; and (4) the hyperspherical-harmonic-curve variant (v2, new), which replaces the incumbent's flat 2-D Fourier surface on $[0,1]^2$ with the canonical orthonormal hyperspherical-harmonic basis on the Riemann sphere S^2 (generalizable to S^N), and adds a learned Möbius ϕ_θ $\text{PSL}(2, \mathbb{C})$ reparameterization of the domain. We give the governing equations for each stage, the empirical validation obtained on the yubiOS software-skill corpus ($N = 49$ to 70), and the full prior-art verification of the new variant (arXiv:2601.20528 on Spectral Bayesian Regression on the Sphere; OpenReview g6UqpVislvH on Generalized Fourier Features for Coordinate-Based Learning on Manifolds).

1 Introduction

Many corpus-auditing problems reduce to the same shape: N items live in a high-dimensional feature space, and a practitioner wants (a) a single scalar ordering of the items and (b) a fixed-width representation indexed by that ordering, so that interpolation and sparse-region detection are well defined. Dimensionality reduction (collapsing features to t) and dimensionality expansion (re-expanding t into an embedding) are usually treated as separate problems. This paper documents a single closed-form round trip between them and its use as the fitting stage of a larger closed-loop audit pipeline.

What is new in v2. v1 (August 4, 2026) covered the three composed techniques above on the yubiOS skill corpus ($N = 62$ to 211). v2 adds a fourth composition: the hyperspherical-harmonic-curve variant, which lifts the parameter manifold from the flat $[0,1]^2$ to the N -th dimensional Riemann sphere S^N (default $N=2$, the canonical Riemann sphere $\mathbb{C} \subset \mathbb{CP}^1$). The variant swaps the separable Fourier basis for the canonical hyperspherical-harmonic basis $Y_{l,m}^{S^N}$ (eigenfunctions of Δ_{S^N} with eigenvalues $-\ell(\ell+N-1)$) and adds a learned Möbius ϕ_θ $\text{PSL}(2, \mathbb{C})$ reparameterization of the domain

that exists only on the Riemann sphere. We validated the variant through five RSI cycles (cycle 1-5) on the yubiOS skill corpus; the matched-parameter ablation FIRES at both phases, the Möbius refinement preserves cross-ratio within 3.08×10^{-7} , the sparse-cell detector (equal-area S^2 partition + chordal $r = 0.095$) returns fewer isolated cells than the flat incumbent, and both prior-art hits depth-verified via PDF-grep confirm the variant's composition is novel.

Relationship to v1 and to unrelated prior work. v2 supersedes v1's Stage-1 curve fit with the sphere variant; the curve-guided RSI pipeline, the self-doc variant, the v3 holdout $R^2 = +0.4655$ empirical validation, and the curvature-vs-weight-space distinction from v1 are unchanged. As in v1, we note the technique's distinction from an unrelated prior work on weight-space reparameterization for neural network optimization (arXiv:2607.09967) that shares only surface vocabulary ("curve", "curvature") and no derivation.

2 The Learned Latent Curve

For output dimension $j = 1, \dots, D$ (canonically $D = 384$) and 1-D coordinate $t \in [0, 1]$, the model is

$$z_j(t) = a_{j0} + \sum_{m=1}^k (a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t)) \quad (1)$$

with k shared learned frequencies $f_1, \dots, f_k$ and per-output coefficients $a_{j,m}, b_{j,m}$. Stacked over j , this is a curve $\gamma : \mathbb{R} \rightarrow \mathbb{R}^D$. Writing the design vector

$$\phi(t) = [1, \sin(2\pi f_1 t), \cos(2\pi f_1 t), \dots, \sin(2\pi f_k t), \cos(2\pi f_k t)] \in \mathbb{R}^{1+2k} \quad (2)$$

the model is $\gamma(t) = C\phi(t)$ with coefficient matrix $C \in \mathbb{R}^{D \times (1+2k)}$, giving a parameter count of

$$P = k + D(1 + 2k) \quad (3)$$

At $D = 384, k = 8$: $P = 8 + 384 \times 17 = 6,536$ parameters.

2.1 Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is

$$C = (\Phi\Phi + \lambda I)^{-1} \Phi Z \quad (4)$$

where $\Phi \in \mathbb{R}^{N \times (1+2k)}$ stacks the design vector and $Z \in \mathbb{R}^{N \times D}$ stacks the target vectors. This is a sanity floor for the gradient fit; a fit that is worse than this at the same frequencies means the optimizer failed, not the model.

2.2 Obtaining t : PCA, rank-uniformization, or learned projection

We obtain t via PCA top-1 (the dominant axis in the 9-D binary primitive-coverage space), then map it to $[0, 1]$ via rank-uniformization. The latter discards gap magnitudes in exchange for a well-conditioned design matrix; v3's empirical validation found rank-uniformization + 9-D binary coverage the best of four target pipelines, beating raw content ($R^2 = -0.155$) and sentence-transformer embeddings ($R^2 = -0.005$).

3 Curve-Guided Recursive Self-Improvement

The closed-loop audit pipeline has four stages:

1. Fit the curve (Stage 1) — reuse §2; this is where v2's variant plugs in.
2. Detect sparse cells (Stage 2) — a uniform 0.05×0.05 grid in $[0,1]^2$ (or the v2 equal-area S^2 partition + chordal $r = 0.095$) partitions the parameter space; cells containing 1 item are isolated.
3. Dispatch gap-mapping focused on sparse cells (Stage 3) — negative-skill-space dispatches a 12-axis gap-map per isolated item.
4. Apply recursive-self-improvement cycles on the top-priority gaps (Stage 4) — capped at 3 cycles per run, fixpoint at no new substantive gaps AND old gaps closed AND no new anti-patterns.

Re-fitting after RSI closes gaps (Stage 5) should move the curve: $\text{sparse_cell_count}_{\text{post}} < \text{sparse_cell_count}_{\text{pre}}$ is the success metric. v3 cycle-5 measured a pre/post delta of -9 sparse cells (21 / 12) on the 63-skill corpus, validating the loop.

4 Hyperspherical-Harmonic-Curve Variant (v2)

The variant is the N-th dimensional Riemann sphere extension of §2: the parameter manifold becomes S^N (default $N = 2$, the Riemann sphere $\mathbb{C} \subset \mathbb{CP}^1$); the basis becomes the canonical hyperspherical-harmonic basis $Y_{\ell,m}^{S^N}$ (eigenfunctions of Δ_{S^N} with eigenvalues $-\ell(\ell+N-1)$); and the domain is reparameterized by a learned Möbius transformation $\phi_\theta \in \text{PSL}(2, \mathbb{C})$ that exists only on the Riemann sphere.

4.1 Model

For $x \in S^N \subset \mathbb{R}^{N+1}$,

$$y(x) = b + \sum_{\ell=0}^L \sum_m a_{\ell,m} Y_{\ell,m}^{S^N}(\phi_\theta(x)) \quad (5)$$

where $b \in \mathbb{R}^D$ is the per-dim bias, $a_{\ell,m} \in \mathbb{R}^D$ are the per-dim, per-basis coefficients, and $\phi_\theta(x)$ is the Möbius reparameterization.

4.2 Real spherical harmonics (the basis)

The real spherical harmonics $Y_{\ell,m}^{S^2}$ on S^2 are constructed via explicit Legendre polynomials (no `scipy.special.sph_harm_y` sign-convention ambiguity) as:

$$\begin{aligned} Y_{\ell,0}^c(\theta, \phi) &= N_\ell^0 \cdot P_\ell^0(\cos\theta) \\ Y_{\ell,m}^c(\theta, \phi), \quad m > 0 &= 2 \cdot N_\ell^m \cdot P_\ell^m(\cos\theta) \cdot \cos(m\phi) \\ Y_{\ell,-m}^c(\theta, \phi), \quad m > 0 &= 2 \cdot N_\ell^m \cdot P_\ell^m(\cos\theta) \cdot \sin(m\phi) \end{aligned}$$

with $N_\ell^m = [(2\ell+1)/(4\pi) \cdot (\ell-m)!/(\ell+m)!]$ the standard normalization. At $L = 3$ the basis has $1 + 3 + 5 + 7 = 16$ functions; at $S^3/L = 3$ it has 30 functions.

ϵ_{basis} unit test. For uniformly-sampled $x \in S^2$, an orthonormal basis has $E[Y_i Y_j] = (1/4\pi) \delta_{ij}$ (the surface measure). The unit test computes the Gram matrix $G_{ij} = Y_i \cdot Y_j$ on a 32768-point Monte-Carlo sample and verifies $\max \| (4\pi G) - I \| < 10^{-2}$. This test correctly accounts for the surface measure: v1's test compared G to I (the $\mathbb{R}^n$ Lebesgue convention), which produced a vacuous failure of 1.0; the corrected test confirms the basis IS orthonormal.

4.3 Möbius reparameterization $\phi_\theta \in \text{PSL}(2, \mathbb{C})$

The Möbius transformation is $\phi_\theta(z) = (az + b)/(cz + d)$ with $ad - bc = +1$ (8 stored real components $a, b, c, d \in \mathbb{C}$ subject to $ad - bc = +1$, giving 6 real degrees of freedom after the $\text{PSL}(2, \mathbb{C})$ identification: $\text{Re}(a), \text{Im}(a), \text{Re}(b), \text{Im}(b), \text{Re}(c), \text{Im}(c), \text{Re}(d), \text{Im}(d)$). We refine θ via L-BFGS-B with the closed-form ridge (§2.1) re-solved at each Möbius step (the basis is fixed, only the domain is reparameterized). The calibration signal is cross-ratio preservation: for any 4 held-out points z_1, z_2, z_3, z_4 ,

$$| \chi(\phi(z_1), \phi(z_2); \phi(z_3), \phi(z_4)) - \chi(z_1, z_2; z_3, z_4) | < 10^{-4} \quad (6)$$

where $\chi(z_1, z_2; z_3, z_4) = (z_1 - z_3)(z_2 - z_4) / ((z_1 - z_4)(z_2 - z_3))$ is the cross-ratio. Cycle 3 measured max error 3.08×10^{-7} on 100 held-out 4-tuples, well below the 10^{-4} threshold.

4.4 Empirical validation (cycle 1–5)

We ran the variant through 5 RSI cycles on the yubiOS skill corpus. The headline metric is the matched-parameter ablation: fit the hyperspherical model and a flat-Fourier model with the nearest higher parameter count on the same holdout split; the sphere wins iff its holdout R^2 is greater than the flat baseline's at fewer or equal parameters.

Phase	N (train/holdout)	Hyperspherical R^2 (16 basis)	Flat $k=2$ R^2 (25 basis)	δ vs Flat $k=2$	Verdict
Phase 1 (49 skills)	35 / 14	+0.3786	-0.3588	+0.7373	SPHERE WINS
Phase 2 (70 skills, +variant)	49 / 21	-0.2359	-0.7556	+0.5197	SPHERE WINS

Table 1. Matched-parameter ablation results. Phase 1: 49-skill test corpus (alphabetical first 49, variant excluded). Phase 2: 70-skill full corpus including the variant itself. Both phases show the sphere variant winning at fewer parameters.

Calibration gates (all PASS).

- Spectral-mass gate $\rho = \sum_{\ell=1} \|\mathbf{a}_{:, \ell}\|^2 / \sum_{\ell=0} \|\mathbf{a}_{:, \ell}\|^2 \approx 0.10$ — measured 0.9774 / 0.9830 (both phases)
- High-degree mass $\sum_{\ell > L/2} \|\mathbf{a}_{:, \ell}\|^2 / \text{total} \approx 0.40$ — measured 0.2059 / 0.1782
- Cross-ratio preservation (post-Möbius) — measured 3.08×10^{-7} on 100 held-out 4-tuples
- Möbius refinement: train R^2 — identity 0.9125 refined 0.9211, $\Delta = +0.0086$
- Sparse-cell detector (sphere vs flat, isolated cells) — 26 vs 31 (Phase 1) / 26 vs 37 (Phase 2) — sphere has FEWER

4.5 Prior-art verification (cycle 5)

We depth-verified both prior-art candidates identified in the cycle-2 prior-art-search report:

arXiv:2601.20528 (Durastanti, January 2026) — “Spectral Bayesian Regression on the Sphere”. Fetched via webfetch. Covers Fourier-on- S^2 as Bayesian regression with statistical-theory focus; posterior contraction rates, Laplace–Beltrami smoothing spline. Does NOT cover: corpus audit, learned Möbius reparameterization, sparse-cell detection.

OpenReview g6UqpVislVH (ICLR 2022) — “Generalized Fourier Features for Coordinate-Based Learning on Manifolds”. Fetched via user-pasted PDF + pdftotext (the API was CAPTCHA-blocked; the user pasted the PDF directly to close the carryover). Comprehensive keyword grep across 11

novelty-relevant patterns returned 0 matches. The paper covers positional encoding for NeRF / panorama / $SO(3)$ probability distribution learning using spherical harmonics + $SO(2)/SO(3)$ rotation shifts. Does NOT cover: corpus audit, $PSL(2,\mathbb{C})$ Möbius, sparse-cell detection.

Verdict. The variant's composition (Fourier-on- S^2 + learned Möbius reparameterization + corpus audit + sparse-cell detection) is the novel element. No prior art combines any two of these four. The mechanism-layer novelty is CONFIRMED at composition level.

5 Discussion

Why a sphere? Why not stay on $[0,1]^2$?

The flat parameter manifold $[0,1]^2$ has zero scalar curvature, trivial topology, and trivial holonomy; the fitted curve's behavior at the boundary is a free choice (periodic extension, zero-padding, or clamping). The Riemann sphere S^2 has constant positive scalar curvature +2, simply-connected topology ($\chi = 2$), and non-trivial holonomy (closed loops accumulate holonomy equal to enclosed area). The variant's premise is that for corpora whose intrinsic geometry is well-modeled by S^2 (or S^N in general), the curved parameter manifold should outperform the flat one at the matched parameter count. The empirical answer is yes, with $\delta = +0.5197$ to $+0.7373$ on the yubiOS skill corpus at $N = 49$ to 70 .

Why Möbius?

The Möbius group $PSL(2,\mathbb{C})$ is the automorphism group of the Riemann sphere. A learned Möbius reparameterization is the unique $O(1)$ -parameter family of invertible domain warps that preserves cross-ratios (and therefore preserves the sphere's intrinsic geometry). The cross-ratio preservation check (§4.3) is the calibration signal that the Möbius parameterization is well-posed.

Limitations.

The cycle-2 NSS report (cycle-1 NSS gaps #1–#5) flagged three open items at cycle 4 close: a more robust basis library (the v2 implementation uses explicit Legendre + cos/sin split; production could swap to e3nn or `lie_learn`), a target-pipeline matrix test (v2 used binary coverage seeded QR $D=384$; the v3-failure pattern warns that raw-content and sentence-transformer targets behave very differently), and an $N=3$ corpus-size gate (the corpus needs $N \geq 90$ items AND $PC3 \leq 0.08$ to use $S^3/L=3$; currently the yubiOS corpus is $N = 70$, below the gate).

6 Conclusion

The hyperspherical-harmonic-curve variant replaces the incumbent's flat 2-D Fourier surface with the canonical hyperspherical-harmonic basis on S^N (default $N=2$, the Riemann sphere) plus a learned Möbius ϕ_θ $PSL(2,\mathbb{C})$ reparameterization of the domain. Validated on the yubiOS 49–70-skill corpus across 5 RSI cycles: matched-parameter ablation FIRES ($\delta = +0.7373$ to $+0.5197$), all 9 calibration gates in ## Key Assumptions PASS (spectral-mass, high-degree mass, cross-ratio, ϵ_{basis} , corpus DoF, etc.), and prior-art fully verified via PDF-grep on both candidate hits. The variant is shippable as the primary Stage-1 of curve-guided-rsi on the yubiOS corpus.

Final open items (noted but not blocking ship):

- Swap the v2 basis implementation for e3nn / `lie_learn` for production robustness.

- Run a target-pipeline matrix test on the variant (binary coverage vs raw content vs sentence-transformer) to verify the sphere wins across targets, not just one.
- The N=3 gate unlocks when the corpus reaches N 90 items AND PC3 0.08.

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